UNIT 1 · TOPIC 3(B)

Tillage Equipment

Syllabus topic: Equipment for land development, tillage, sowing, planting and transplanting

Tillage — definitions

Tillage is the mechanical manipulation of soil to give the crop a favourable environment to grow.

Type	Depth / nature	Implements
Primary tillage	Initial, deeper opening — 12–30 cm or more	MB plough, disc plough, chisel plough, subsoiler, rotary plough
Secondary tillage	Shallower, refining — 5–10 cm	Harrows, cultivators, levellers, pulverizers, rollers

- **Minimum tillage:** the least soil manipulation needed.
- **Zero/no-till:** direct seeding into unprepared soil.
- **Conservation tillage: ≥30%** of the surface stays covered by residue after planting.
- Strip, rotary and mulch tillage are other classes.

Mouldboard (MB) plough

The MB plough cuts, lifts, inverts and pulverizes the furrow slice. **Complete inversion** is its distinguishing action, so it buries weeds and trash well.

Part	Function
Share	Penetrates and cuts.
Mouldboard	Lifts and inverts.
Landside	Absorbs side thrust; its rear end is the heel.
Frog	Base to which all parts bolt.
Shin	Cutting-edge extension of the mouldboard.
Tail piece	—

- **Share types:** slip share, slip-nose share, shin share, bar-point share (most common where abrasive soils wear points fast; the point can be extended).
- **Share materials:** chilled cast iron, or soft-centre steel (hard outer layers over a soft core).

Mouldboard type	Shape / use
General purpose	Medium pitch — most common.
Stubble	Short, abruptly curved — maximum pulverization.
Sod / breaker	Long, gently curved — complete inversion without breaking the slice.
Slat	Sticky soils.
High-speed	Slender — for 6–8 km/h+.

- **Vertical (down) suction** helps penetration — typically **3–5 mm** on tractor ploughs.
- **Horizontal (side/land) suction** helps maintain furrow width — typically **3–10 mm**.
- **Plough size** = width of cut of one bottom (e.g., 2 × 30 cm = 2 bottoms of 30 cm each).
- **Unit draft** = draft ÷ furrow cross-section: light sandy soils ~0.2–0.4 kgf/cm², heavy clays 0.7–1.4 kgf/cm²; draft varies with speed as $D = a + bV^2$ (Kepner). See Topic 2(a).

Ploughing method	How it works
Gathering	Plough clockwise, starting at the centre.
Casting	Plough anticlockwise, starting at the sides.
Continuous	Use a one-way/reversible plough — no back furrows or dead furrows; ideal for hillsides and irrigated land.

Disc plough

The disc plough cuts, partially inverts and mixes the soil. A standard disc plough has **no landside** — the rear furrow wheel takes the side thrust.

- **Standard disc plough:** individually mounted concave discs, **60–90 cm** diameter, **6–10 mm** thick.
- **Disc angle** (between the disc face plane and the direction of travel): **42–45°**. A larger disc angle widens the cut and improves penetration in hard soil.
- **Tilt angle** (between the disc plane and the vertical): **15–25°**. It increases penetration in hard soils.
- **Penetration comes from weight** (the MB plough uses suction instead), so discs carry added ballast — typically **180–540 kg** per disc bottom equivalent.
- **Vertical disc plough** (harrow plough / one-way disc): discs on a common gang axle, **50–60 cm** discs, shallow (**8–10 cm**) — the "wheatland plough".

Advantages over the MB plough	Disadvantages
Works in sticky, hard, stony and rooty soils, and where scouring is a problem; rolls over obstacles.	Incomplete inversion; leaves a cloddy surface; needs high weight; needs a rear furrow wheel.

Chisel plough, subsoiler and rotavator

- **Chisel plough:** narrow curved shanks break hard pan/soil to **30–45 cm** without inversion; shanks typically spaced **30 cm** apart.
- **Subsoiler:** a heavier standard that shatters impervious hardpan at **45–90 cm**; needs **60–100 hp**. Winged subsoilers widen the shatter zone. A **mole plough** forms drainage channels (mole drains) in clay soils.
- **Rotavator (rotary tiller):** PTO-driven blades on a horizontal shaft — the **L-type** is most common in India for trashy conditions (C and J types also exist). It cuts and pulverizes in a single pass, giving a seedbed in one operation. It can produce negative draft. **Bite length** (forward travel per blade strike) determines pulverization. Power consumption is high; it is excellent for puddling.
- Rotavator blade tip speed **4–9 m/s**; rotor speed **200–350 rpm** typical.
- **Puddling:** break soil aggregates in standing water to form an impervious layer for paddy — use a cage wheel + rotavator, or a peg-type puddler.

Secondary tillage

- **Disc harrow:** gangs of concave discs (**35–70 cm**). Types: single action, tandem (double action), offset (works offset to the tractor centre-line — orchards), plough harrow. Gang angle **15–30°**. Increase penetration by increasing the gang angle or adding weight.
- **Cultivators:** rigid tine; spring-loaded tine (stony soils — trips and resets); duck-foot (sweep) cultivator for shallow weed kill and moisture conservation.
- **Blade harrow (bakhar):** indigenous implement — cuts weeds below the surface without inversion and retains the residue mulch.
- **Rollers/pulverizers:** clod crushers and plankers compact and firm the seedbed.

Practice MCQs

Q9. Complete inversion of the furrow slice is the distinguishing action of:

(a) Disc plough
(b) Chisel plough
(c) Mouldboard plough
(d) Cultivator

Q10. The part of the MB plough that absorbs side thrust is the:

(a) Frog
(b) Share
(c) Landside
(d) Shin

Q11. Bar-point share is preferred in:

(a) Sticky soils
(b) Abrasive/stony soils causing rapid point wear
(c) Sandy loams
(d) Peat soils

Q12. A stubble-type mouldboard is characterized by:

(a) Long, gently curved surface
(b) Short, abruptly curved surface for maximum pulverization
(c) Slats
(d) Very slender shape for high speed

Q13. Vertical suction in an MB plough helps in:

(a) Maintaining furrow width
(b) Penetration to proper depth
(c) Inversion
(d) Reducing draft

Q14. In a standard disc plough, the disc angle usually ranges between:

(a) 15–25°
(b) 30–35°
(c) 42–45°
(d) 50–60°

Q15. Penetration in a disc plough is obtained mainly by:

(a) Suction
(b) Weight of the plough
(c) Hydraulic down-pressure only
(d) Sharpness of share

Q16. Which plough has no landside?

(a) MB plough
(b) Standard disc plough
(c) Turn-wrest plough
(d) Victory plough

Q17. Continuous ploughing without back furrows and dead furrows is achieved with:

(a) Gathering method
(b) Casting method
(c) Two-way (reversible) plough
(d) Tandem harrow

Q18. A subsoiler normally operates at a depth of:

(a) 10–15 cm
(b) 20–30 cm
(c) 45–90 cm
(d) 100–150 cm

Q19. The blade type most common on Indian rotavators for trashy conditions is:

(a) C-type
(b) J-type
(c) L-type
(d) Pick type

Q20. An offset disc harrow is most suited for:

(a) Puddling
(b) Orchards — working under tree canopies
(c) Deep tillage
(d) Ridging

Answers

Q9 — (c) Mouldboard plough

Q10 — (c) Landside

Q11 — (b) Abrasive/stony soils causing rapid point wear

Q12 — (b) Short, abruptly curved surface for maximum pulverization

Q13 — (b) Penetration to proper depth — *MB plough penetrates by suction; disc plough penetrates by weight — a classic contrast question.*

Q14 — (c) 42–45°

Q15 — (b) Weight of the plough — *MB plough penetrates by suction; disc plough penetrates by weight — a classic contrast question.*

Q16 — (b) Standard disc plough

Q17 — (c) Two-way (reversible) plough

Q18 — (c) 45–90 cm

Q19 — (c) L-type

Q20 — (b) Orchards — working under tree canopies